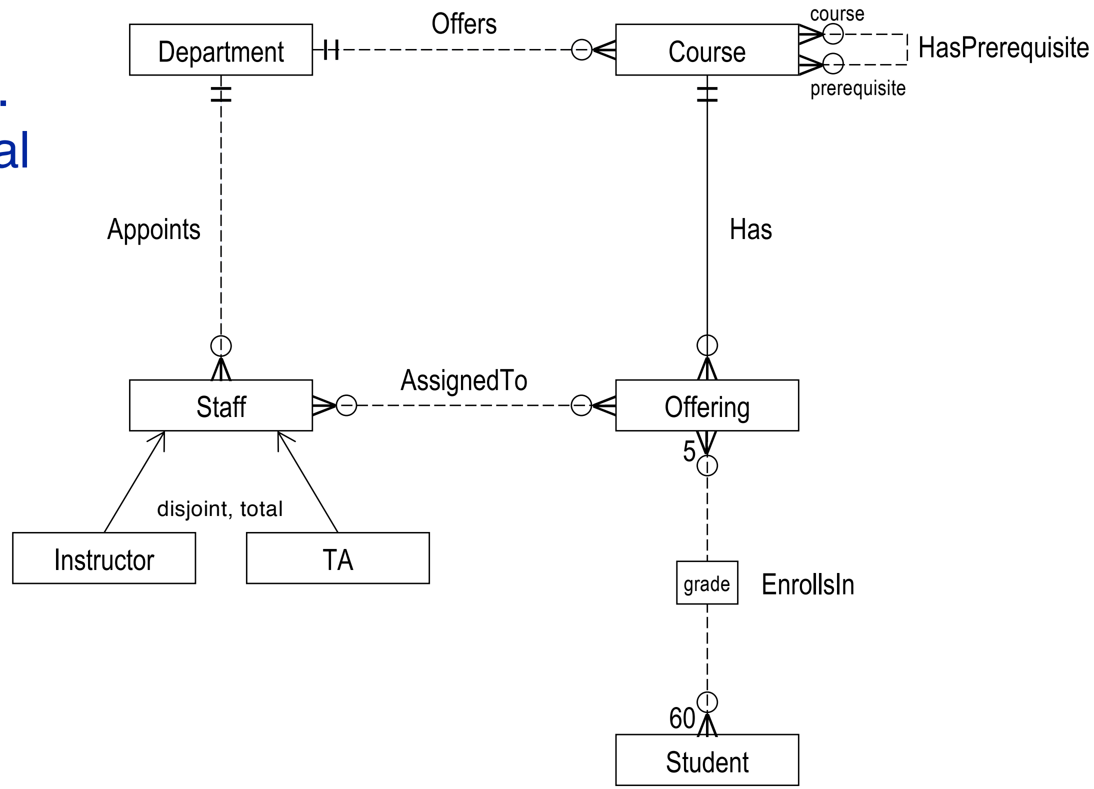
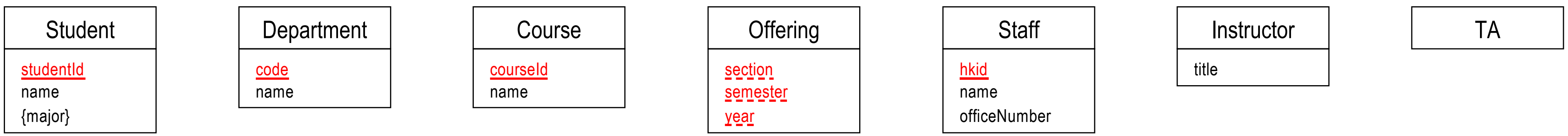
COMP 3311
DATABASE MANAGEMENT SYSTEMS

LECTURE 4 EXERCISES
RELATIONAL DATABASE DESIGN:
E-R SCHEMA TO RELATION SCHEMAS REDUCTION

EXERCISE 1:
UNIVERSITY E-R SCHEMA REDUCTION

Reduce the university E-R schema to relation schemas. Specify all keys and referential integrity constraints.





EXERCISE 1: REDUCE STRONG ENTITIES

Reduce **Student**, **Course** and **Department** strong entities

Student
studentId name {major}

Student(studentId, name)

Course
courseId name

Course(courseId, name)

Department
code name

Department(code, name)

How do we reduce the strong entities?

⟹ Create a relation for each strong entity with the same attributes as the entity (*excluding* multivalued and composite attributes).

What are the keys of these relations? ⟹ Same as the entities.

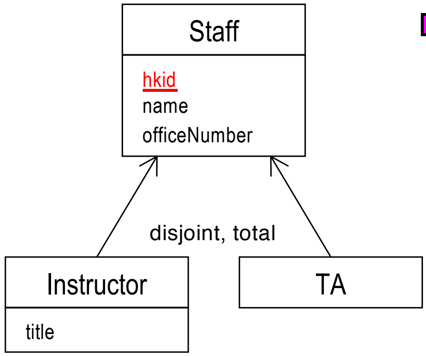
What are the foreign key constraints? ⟹ None.

EXERCISE 1: REDUCE GENERALIZATIONS

Reduce **Staff** generalization

Which option to select?

Option 1: Reduce *all entities* to relation schemas.

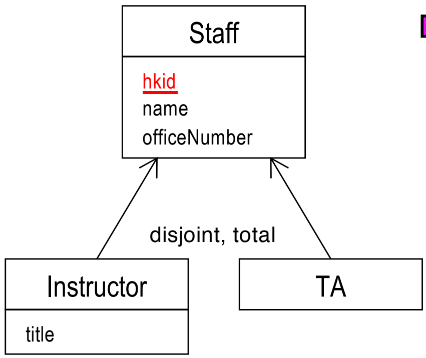


Staff(hkid, name, officeNumber)

Instructor(*hkid*, title)
foreign key (hkid) references Staff(hkid)
on delete cascade

TA(*hkid*)
foreign key (hkid) references Staff(hkid)
on delete cascade

Option 2: Reduce *only subtype entities* to relation schemas.



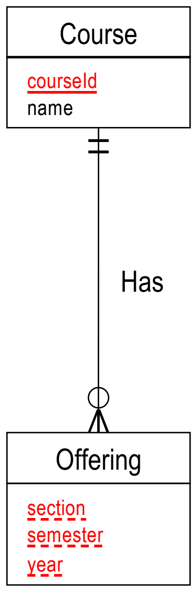
Instructor(hkid, name, officeNumber, title)

TA(hkid, name, officeNumber)

Select Option 1 since only **Staff** has a relationship to other entities and some subtype entities have their own attributes.

EXERCISE 1: REDUCE WEAK ENTITIES

Reduce **Offering** weak entity together with **Has** identifying relationship



Course(courseId, name) *(previously reduced)*

Offering(*courseId*, section, semester, year)
foreign key (courseId) references Course(courseId)
on delete cascade

SCHEMA REDUCTION NOTE
Since the key for **Offering** contains several attributes, to simplify the subsequent reductions that involve **Offering** and to facilitate relation implementation, it would be helpful to introduce a surrogate key for **Offering** such as offeringId.
(Not done here.)

How do we reduce this entity?

⟹ Create a relation from **Offering** and include courseId, the key for **Course**, as a foreign key.

What is the key for this relation?

What is the foreign key constraint?

What is the referential integrity action?

EXERCISE 1:
REDUCE COMPOSITE/MULTIVALUED ATTRIBUTES

Reduce major multivalued attribute

Student(studentId, name)
(previously reduced)

Student
studentId name {major}

StudentMajor(*studentId, major*)
foreign key (studentId) references Student(studentId)
on delete cascade

How do we reduce the multivalued attribute major?

⟹ Create a relation **StudentMajor** and include studentId, the key for **Student**, and the attribute major.

What is the key for this relation?

What is the foreign key constraint?

What is the referential integrity action—on delete cascade or on delete set null?

EXERCISE 1: REDUCE RELATIONSHIPS

Reduce **Offers** 1:N relationship between **Department** and **Course**

Department(<u>code</u>, name) *(previously reduced)*   Course(<u>courseId</u>, name) *(previously reduced)*

Offers(*code*, *<u>courseId</u>*)
  foreign key (code) references Department(code)
    on delete cascade
  foreign key (courseId) references Course(courseId)
    on delete cascade

How do we reduce this relationship?

⟹ Create a relation, **Offers**, containing the keys for **Department** and **Course**.

What is the key for the relation?

What are the foreign key constraints?

What are the referential integrity actions?

- *1:1 relationship* ⟹ the primary key for either entity.
- *1:N relationship* ⟹ the primary key for the entity on the N-side of the relationship.
- *N:M relationship* ⟹ the union of the primary keys for the two participating entities.

EXERCISE 1: REDUCE RELATIONSHIPS

Reduce **Offers** 1:N relationship between **Department** and **Course**
(using schema combination)

Department(<u>code</u>, name) *(previously reduced)*

Course(<u>courseId</u>, name, *code*)
foreign key (code) references Department(code)
on delete cascade

<u>SCHEMA REDUCTION NOTE</u>
The referential integrity action for the relation into which the foreign key is placed is determined by the participation constraint of the reduced entity.
- partial: on delete set null
- total: on delete cascade

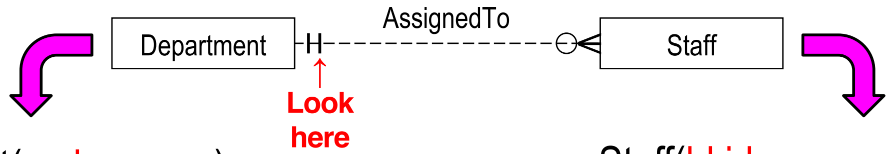
Which relation do we use?
⟹ **Course** (Add code, the key for **Department**, as a foreign key.)

What is the foreign key constraint?

What is the referential integrity action?

EXERCISE 1: REDUCE RELATIONSHIPS

Reduce **Appoints** 1:N relationship between **Department** and **Staff**
(using schema combination)



Department(code, name) *(previously reduced)*

Staff(hkid, name, officeNumber, *code*)
foreign key (code) references Department(code)
on delete cascade

Which relation do we use?

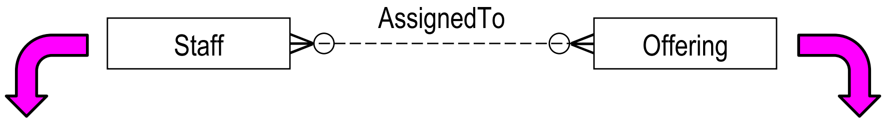
⟹ **Staff** (Add code, the key for **Department**, as a foreign key.)

What is the foreign key constraint?

What is the referential integrity action?

EXERCISE 1: REDUCE RELATIONSHIPS

Reduce **AssignedTo** N:M relationship between **Staff** and **Offering**



Staff(hkid, name, officeNumber)
(previously reduced)

Offering(*courseId*, section, semester, year)
(previously reduced)

AssignedTo(*hkid, courseId, section, semester, year*)
foreign key (hkid) references Staff
on delete cascade
foreign key (courseId, section, semester, year) references Offering
on delete cascade

How do we reduce this relationship?

⟹ Create a relation, **AssignedTo**, containing the keys for the **Staff** and **Offering** relations.

What is the key for the relation?

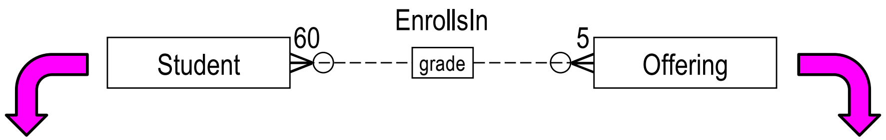
What are the foreign key constraints?

What are the referential integrity actions?

SCHEMA REDUCTION NOTE
For a relation that represents a relationship, the referential integrity action is always on delete cascade.

EXERCISE 1: REDUCE RELATIONSHIPS

Reduce **EnrollsIn** N:M relationship between **Student** and **Offering**



Student(<u>studentId</u>, name)
(previously reduced)

Offering(<u>*courseId, section, semester, year*</u>)
(previously reduced)

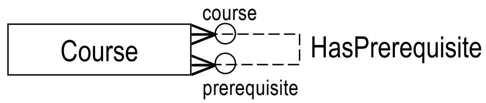
EnrollsIn(<u>*studentId, courseId, section, semester, year*</u>, grade)
foreign key (studentId) references Student(studentId)
on delete cascade
foreign key (courseId, section, semester, year) references
Offering(courseId, section, semester, year)
on delete cascade

How do we reduce this relationship?
⟹ Create a relation, **EnrollsIn**, containing the keys for the **Student** and **Offering** relations.

Anything else? ⟹ Add the attribute grade to the relation.

EXERCISE 1: REDUCE RELATIONSHIPS

Reduce **HasPrerequisite** N:M relationship between **Course** and **Course**



Course(<u>courseId</u>, name) *(previously reduced)*

HasPrerequisite(<u>*courseId, prerequisiteId*</u>)
foreign key (courseId) references Course(courseId)
on delete cascade
foreign key (prerequisiteId) references Course(courseId)
on delete cascade

How do we reduce this relationship?

⟹ Create a relation, **HasPrerequisite**, containing the key for the **Course** relation (twice).

What is the key for the relation?

EXERCISE 1: UNIVERSITY RELATION SCHEMAS

Staff(hkid, name, officeNumber, *code*)
  foreign key (code) references Department(code)
    on delete cascade

Instructor(*hkid*, title)
  foreign key (hkid) references Staff(hkid)
    on delete cascade

TA(*hkid*)
  foreign key (hkid) references Staff(hkid)
    on delete cascade

Student(studentId, name)

Course(courseId, name, *code*)
  foreign key (code) references Department(code)
    on delete cascade

Department(code, name)

StudentMajor(*studentId*, major)
  foreign key (studentId) references Student(studentId)
    on delete cascade

Offering(*courseId*, section, semester, year)
  foreign key (courseId) references Course(courseId)
    on delete cascade

AssignedTo(*hkid, courseId, section, semester, year*)
  foreign key (hkid) references Staff(hkid)
    on delete cascade
  foreign key (courseId, section, semester, year) references
    Offering(courseId, section, semester, year)
    on delete cascade

EnrollsIn(*studentId, courseId, section, semester, year*, grade)
  foreign key (studentId) references Student(studentId)
    on delete cascade
  foreign key (courseId, section, semester, year) references
    Offering(courseId, section, semester, year)
    on delete cascade

HasPrerequisite(*courseId, prerequisiteId*)
  foreign key (courseId) references Course(courseId)
    on delete cascade
  foreign key (prerequisiteId) references Course(courseId)
    on delete cascade